

TANMAY SHARMA

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Education

Northeastern University

Master of Science in Cybersecurity - 3.70/4.00

September 2024 – Present

Boston, MA

Coursework: Network Security, Linux Administration, Cryptography, Binary Exploitation, Linux Rootkit Development, Shell Scripting, Security Risk Management and Assessment

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering - 3.56/4.00

September 2020 – July 2024

Chennai, India

Coursework: Operating Systems, Computer Architecture, Data Structures, Algorithms, Computer Networks

Experience

Khoury School of Computer Science

Teaching Assistant

January 2026 – Present

Boston, MA

- Conducting weekly lab sessions for 50+ students on network security, system security, Docker, and Linux administration topics, providing hands-on guidance and real-time debugging support to ensure students grasp complex technical concepts and complete practical assignments
- Grading and providing constructive feedback on programming assignments and security implementation projects, maintaining consistent evaluation standards while helping students identify areas for improvement in their system design and security practices
- Assisting professor in developing supplementary course materials including lab documentation, Docker environment setup guides, etc. Also did QA and testing for Viceroy Decree's Sentinel CTF 2026.

CyberspACe securiTy and forensIcs lab (CactiLab)

Research Volunteer

March 2025 – August 2025

Boston, MA

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Technical Skills

Languages: C, C++, Python, SQL, Bash, PowerShell

Security Tools: Burp Suite, Metasploit Framework, Nessus, Wireshark, Nmap, Ghidra, Static Analysis Tools

Protocols & Standards: OWASP Top 10, CWE, MITRE ATT&CK, HTTP/HTTPS, SSL/TLS, OAuth, TCP/IP

Technologies: Vulnerability Analysis, Secure Code Reviews, Active Directory, Docker, Git, Linux Administration

Projects

Active Directory Lab | AD, LLMNR, Bloodhound, Kerberoasting

November 2025

- Designed and deployed a complete Active Directory pentesting environment with Windows Server 2022 Domain Controller and 2 Windows 10 Enterprise clients, configuring certificate services, domain forests, and realistic user scenarios to simulate enterprise infrastructure vulnerabilities
- Executed comprehensive security assessment covering 15+ attack vectors including LLMNR poisoning, SMB relay, Kerberoasting, and Golden Ticket attacks, documenting exploitation methodologies and defense strategies through systematic enumeration with BloodHound, ldapdomaindump, and PingCastle

Secure Instant Messaging System | Python, SRP, ECDH, TCP Sockets

January 2025 – April 2025

- Built a secure terminal-based chat system using SRP, ECDH, and AES-GCM for confidential, authenticated real-time communication (CY6740 - Network Security).
- Developed socket-based architecture with encrypted commands, rate limiting, and user listing, cutting abuse vectors by 80% and boosting resilience against attacks.

OSINT-Recon LLM Framework | Python, Mistral 7B, FAISS

October 2025 – Present

- Developed an AI-powered OSINT reconnaissance framework integrating Mistral 7B LLM with automated subdomain enumeration, DNS resolution, and HTTP service discovery tools for intelligent security analysis.
- Implemented RAG (Retrieval-Augmented Generation) system using FAISS vector database and sentence transformers for semantic search, with asynchronous tool orchestration generating automated threat intelligence reports.

Certifications, Achievements & Community

Certifications: [CompTIA Security+](#) | [AWS Academy Cloud Security Foundations](#) | PNPT by TCM Security (In Progress)

Achievements: [SimSpace Cyber Cup 2025 - Rank 6](#), Good rank in numerous CTFs

Community: Member of Boston Mesh Project and Boston Open Dev Community